UNIVERSITY OF PERADENIYA
FACULTY OF SCIENCE

SEMESTER II - 2022/2023

MT 105 / MAT 1023 - REAL ANALYSIS I

END SEMESTER EXAMINATION

Time allowed: Three Hours **Answer ALL FOUR questions**

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1. (a) For the statements i. and ii. below:

- Determine whether each of the statement is **True** or **False**.
- If the statement is **True**, provide a short proof to support it.
- If the statement is **False**, provide a counterexample with a clear justification to disprove it.

i. If $A + B := \{a + b : a \in A \text{ and } b \in B\}$, where A and B are nonempty bounded subsets of $\mathbb{R}$, then $\sup(A + B) = \sup(A) + \sup(B)$.

ii. If $A - B := \{a - b : a \in A \text{ and } b \in B\}$, where A and B are nonempty bounded subsets of $\mathbb{R}$, then $\sup(A - B) = \sup(A) - \sup(B)$.

(40 points)

(b) Find the supermum and the infimum of the following sets:

i. $A = \left\{ \dfrac{m}{n} + \dfrac{4n}{m} \ : \ m, n \in \mathbb{Z}^+ \right\}$ (**Hint :** For $a, b > 0$, $\dfrac{a+b}{2} \geq \sqrt{ab}$)

ii. $B = \left\{ \dfrac{mn}{4m^2 + n^2} \ : \ m \in \mathbb{Z}, n \in \mathbb{Z}^+ \right\}$ (**Hint :** Use i.)

(40 points)

(c) Let $r < 0$ be a negative real number. Use the *Archimedean principle* (or the consequences of the *Archimedean principle* we proved in class) to prove that there is an $n \in \mathbb{Z}^+$ such that $r < -1/n < 0$. **(20 points)**

2. (a) For the statements i. and ii. below:

- Determine whether each of the statement is **True** or **False**.
- If the statement is **True**, provide a short proof to support it.
- If the statement is **False**, provide a counterexample with a clear justification to disprove it.

i. If $x_n \to \infty$ as $n \to \infty$, then $\dfrac{1}{x_n} \to 0$ as $n \to \infty$.

ii. If $x_n \to 0$ as $n \to \infty$, then $\dfrac{1}{x_n} \to \infty$ as $n \to \infty$.

(40 points)

[Turn over...

(b) Given that $e = \lim_{n\to\infty}\left(1+\frac{1}{n}\right)^n$ (where e is the *Euler's number*), explain why the limit of $\left\{\left(1+\frac{1}{n^2}\right)^{2n^2}\right\}$ exists and find the limit (in terms of e). **(20 points)**

(c) State the *Monotone Convergence Theorem* (MCT).

Define a sequence $\{x_n\}$ by

$$x_1 = \sqrt{3}, \quad x_2 = \sqrt{3+\sqrt{3}}, \quad x_{n+1} = \sqrt{3+x_n} \text{ for } n \geq 2.$$

i. Use Mathematical Induction on n to show that $\{x_n\}$ is bounded above by 3.

ii. Use Mathematical Induction on n to show that $\{x_n\}$ is increasing.

iii. Using i and ii, find $\lim_{n\to\infty} x_n$. **(55 points)**

3. (a) Use the precise $\epsilon - \delta$ definition to prove that $\lim_{x\to 1}\frac{2x^2-3x+1}{x-1} = 1$. **(20 points)**

(b) State the *Extreme Value Theorem* (EVT) for a closed and bounded interval.

Provide an example for each of the following cases: The conclusion of the EVT fails to hold when the interval is

i. not closed

ii. not bounded

(25 points)

(c) Consider the *Thomae's function* defined by

$$f(x) = \begin{cases} \frac{1}{q} & : x = \frac{p}{q} \in \mathbb{Q} \text{ (in reduced form)}, \\ 0 & : x \notin \mathbb{Q}. \end{cases}$$

i. Prove that f is discontinuous at every rational number in the interval $(0,1)$.

ii. Prove that f is continuous at every irrational number in the interval $(0,1)$.

(30 points)

(d) State the *Intermediate Value Theorem* (IVT).

Assume that $f, g : [a,b] \to \mathbb{R}$ are continuous and such that $f(a) < g(a)$ and $f(b) > g(b)$. Prove that there exists $x_0 \in (a,b)$ for which $f(x_0) = g(x_0)$.

(20 points)

4. (a) Assume that $f(0) = 0$ and that f is differentiable at zero. For $k \in \mathbb{Z}^+$, show that

$$\lim_{x \to 0} \frac{1}{x} \left(f(x) + f\left(\frac{x}{2}\right) + f\left(\frac{x}{3}\right) + \cdots + f\left(\frac{x}{k}\right) \right) = \left(1 + \frac{1}{2} + \frac{1}{3} + \cdots + \frac{1}{k}\right) f'(0).$$

(20 points)

(b) For i. and ii. below, determine whether *Rolle's Theorem* can be applied to the function on the closed interval $[a, b]$.

If *Rolle's Theorem* can be applied, find all values of $c \in (a, b)$ such that $f'(c) = 0$. If Rolle's Theorem cannot be applied, explain why.

i. $f(x) = (x + 4)^2(x - 3)$ on $[-4, 3]$

ii. $f(x) = 3 - |x - 3|$ on $[0, 6]$

(30 points)

(c) Prove that $1 + x < e^x$ for all $x > 0$. **(20 points)**

(d) Suppose that f is differentiable on (a, b), continuous on $[a, b]$, and that $f(a) = f(b) = 0$. Prove that if $f(c) \neq 0$ for some $c \in (a, b)$, then there exist $x_1, x_2 \in (a, b)$ such that $f'(x_1)$ is positive and $f'(x_2)$ is negative. **(20 points)**

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